



**Department of Computer Science and Engineering (Data Science)**  
**B.Tech. Sem: III Subject: Statistics for Data Science**  
**Experiment 8**

**Name: Bhuvi Ghosh**

**SAPID: 60009210191**

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Date:    | Experiment Title: Chisquare Test using Python                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Aim      | To study test of independence of attributes and goodness of fit.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Software | Google Colab                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Theory   | <p><b>Question 1:</b> For a give Titanic dataset, can it be concluded that gender and survival of passengers are related to each other?</p> <pre> import numpy as np import pandas as pd import math import scipy import scipy.stats as stats import statistics as st  from scipy.stats import chi2  def chi2_ind_att(df,alpha):     row_total=sum(df.values.T)     column_total=sum(df.values)     total=sum(row_total)     obs_freq=[]     exp_freq=[]     chi2_stats=0     m=len(row_total)     n=len(column_total)     for i in range(m):         for j in range (n):             exp=round(row_total[i]*column_total[j]/total)             exp_freq.append(exp)             obs=df.values[i][j]             obs_freq.append(obs)             chi2_stats=chi2_stats+(obs-exp)**2/exp  print("Observed Frequency = ",obs_freq) print("Total Observed Frequency= "+str(sum(obs_freq))) print("Expecteed Frequency = ",exp_freq) print("Total Expecteed Frequency= "+str(sum(exp_freq))) dof=(m-1)*(n-1) p_value=chi2.sf(chi2_stats,dof) print("chisquare_statistics=",chi2_stats,'and p_value=',p_value) if p_value&gt;alpha:     print("Failed to reject null hypothesis for level of significance= "+str(alpha)) else:     print("NULL hypothesis is rejected for level of significance= "+str(alpha)) </pre> |



```
from google.colab import drive
drive.mount('/content/drive')
```

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount(e).

```
path = "/content/drive/MyDrive/SDS dataset/Titanic.csv"
data = pd.read_csv(path)
data
```

|      | Name                                          | PClass | Age   | Sex    | Survived |
|------|-----------------------------------------------|--------|-------|--------|----------|
| 0    | Allen, Miss Elisabeth Walton                  | 1st    | 29.00 | female | 1        |
| 1    | Allison, Miss Helen Loraine                   | 1st    | 2.00  | female | 0        |
| 2    | Allison, Mr Hudson Joshua Creighton           | 1st    | 30.00 | male   | 0        |
| 3    | Allison, Mrs Hudson JC (Bessie Waldo Daniels) | 1st    | 25.00 | female | 0        |
| 4    | Allison, Master Hudson Trevor                 | 1st    | 0.92  | male   | 1        |
| ...  | ...                                           | ...    | ...   | ...    | ...      |
| 1308 | Zakarian, Mr Artun                            | 3rd    | 27.00 | male   | 0        |
| 1309 | Zakarian, Mr Maprieder                        | 3rd    | 26.00 | male   | 0        |
| 1310 | Zenni, Mr Philip                              | 3rd    | 22.00 | male   | 0        |
| 1311 | Lievens, Mr Rene                              | 3rd    | 24.00 | male   | 0        |
| 1312 | Zimmerman, Leo                                | 3rd    | 29.00 | male   | 0        |

```
df=pd.crosstab(index=data['Sex'], columns=data['Survived'])
df
```

|          |     |     |
|----------|-----|-----|
| Survived | 0   | 1   |
| Sex      |     |     |
| female   | 154 | 308 |
| male     | 709 | 142 |

```
print("Null Hypothesis is that gender and survival of passengers are independent of each other")
print("Alternate Hypothesis is that gender and survival of passengers are related to each other")
alpha=0.05
chi2_ind_att(df, alpha)
```

Null Hypothesis is that gender and survival of passengers are independent of each other  
Alternate Hypothesis is that gender and survival of passengers are related to each other  
Observed Frequency = [154, 308, 709, 142]  
Total Observed Frequency= 1313  
Expected Frequency = [304, 158, 559, 292]  
Total Expected Frequency= 1313  
chisquare\_statistics= 333.72346293361545 and p\_value= 1.4853394683004594e-74  
Null hypothesis is rejected for level of significance= 0.05

**Question 2:** For a give Titanic dataset, can it be concluded that class and survival of passengers are related to each other?

```
df=pd.crosstab(index=data['PClass'], columns=data['Survived'])
df
```

|          |     |     |
|----------|-----|-----|
| Survived | 0   | 1   |
| PClass   |     |     |
| 1st      | 129 | 193 |
| 2nd      | 160 | 119 |
| 3rd      | 573 | 138 |

```
print("Null Hypothesis is that gender and survival of passengers are independent of each other")
print("Alternate Hypothesis is that gender and survival of passengers are related to each other")
alpha=0.05
chi2_ind_att(df, alpha)
```

Null Hypothesis is that gender and survival of passengers are independent of each other  
Alternate Hypothesis is that gender and survival of passengers are related to each other  
Observed Frequency = [129, 193, 160, 119, 573, 138]  
Total Observed Frequency= 1312  
Expected Frequency = [212, 110, 183, 96, 467, 244]  
Total Expected Frequency= 1312  
chisquare\_statistics= 173.63282029663736 and p\_value= 1.9774801554824393e-38  
Null hypothesis is rejected for level of significance= 0.05

**Question 3:** A table shows the number of men and women buying different types of pets. Can it be concluded that gender and choice of pet are related to each other?

|       | dog | cat | bird | total |
|-------|-----|-----|------|-------|
| men   | 207 | 282 | 241  | 730   |
| women | 234 | 242 | 232  | 708   |
| total | 441 | 524 | 473  | 1438  |

```
data=[[207,282,241],[234,242,232]]
df=pd.DataFrame(data, columns=['dog','cat','bird'], index=['men','women'])
print(df)
```

```
      dog  cat  bird
men    207  282  241
women  234  242  232
```

```
print("Null Hypothesis is that gender and choice of pet are independent of each other")
print("Alternate Hypothesis is that gender and choice of pet are related to each other")
alpha=0.05
chi2_ind_att(df, alpha)
```

Null Hypothesis is that gender and choice of pet are independent of each other  
 Alternate Hypothesis is that gender and choice of pet are related to each other  
 Observed Frequency = [207, 282, 241, 234, 242, 232]  
 Total Observed Frequency= 1438  
 Expected Frequency = [224, 266, 240, 217, 258, 233]  
 Total Expected Frequency= 1438  
 chisquare\_statistics= 4.58508839566494 and p\_value= 0.1010091474093573  
 Failed to reject null hypothesis for level of significance= 0.05

**Question 4:** For the given drug data, can we conclude that treatment is effective?

```
data=[[60,10],[30,25]]
df=pd.DataFrame(data, columns=['Cured','Non-Cured'], index=['Treatment','Non-Treatment'])
print(df)
```

```
      Cured  Non-Cured
Treatment    60         10
Non-Treatment  30         25
```

```
print("Null Hypothesis is that treatment is effective.")
print("Alternate Hypothesis is that treatment is not effective.")
alpha=0.05
chi2_ind_att(df, alpha)
```

Null Hypothesis is that treatment is effective.  
 Alternate Hypothesis is that treatment is not effective.  
 Observed Frequency = [60, 10, 30, 25]  
 Total Observed Frequency= 125  
 Expected Frequency = [50, 20, 40, 15]  
 Total Expected Frequency= 125  
 chisquare\_statistics= 16.166666666666668 and p\_value= 5.800591546183077e-05  
 Null hypothesis is rejected for level of significance= 0.05

**Question 5:** The table below is an exit poll which displays the joint responses to 2 categorical variables: people in categories from 18–29, 30–



44, 45–64 and >65 years, and their political affiliation, which is “Conservative”, “Socialist” and “Other”. Create data corresponding to this information. Is there any evidence of a relationship between the age group and their political affiliation, at 5% significant level?

|            | Conservative | Socialist | Other | Total |
|------------|--------------|-----------|-------|-------|
| 18-29      | 141          | 68        | 4     | 213   |
| 30-44      | 179          | 159       | 7     | 345   |
| 45-64      | 220          | 216       | 4     | 440   |
| 65 & older | 86           | 101       | 4     | 191   |
| Total      | 626          | 544       | 19    | 1189  |

```
data=[[141,68,4],[179,159,7],[220,216,4],[86,101,4]]
df=pd.DataFrame(data, columns=['Conservative','Socialist','Other'], index=['18-29','30-44','45-64','65+'])
print(df)
```

```
Conservative  Socialist  Other
18-29         141        68    4
30-44         179       159    7
45-64         220       216    4
65+           86       101    4
```

```
print("Null Hypothesis is that there is no relationship between the age group and their political affiliation.
print("Alternate Hypothesis is that there is a relationship between the age group and their political affiliation.
alpha=0.05
chi2_ind_att(df, alpha)
```

```
Null Hypothesis is that there is no relationship between the age group and their political affiliation.
Alternate Hypothesis is that there is a relationship between the age group and their political affiliation.
Observed Frequency = [141, 68, 4, 179, 159, 7, 220, 216, 4, 86, 101, 4]
Total Observed Frequency= 1189
Expected Frequency = [112, 97, 3, 182, 158, 6, 232, 201, 7, 101, 87, 3]
Total Expected Frequency= 1189
chisquare_statistics= 24.57454792237695 and p_value= 0.0004092601285044903
Null hypothesis is rejected for level of significance= 0.05
```

**Question 6:** A researcher takes a random sample and pick 123 students about their party affiliation. Out of them 57 vote for party A, 26 vote for party B and 40 for Others. Generally, 41.5% of people vote for the party A, 25.7% for the party B and the remaining 32.8% as Others. Test the hypothesis that sample data follows given distribution.

```
Obs_Votes=np.array([57,26,40])
Exp_Votes_percent=np.array([41.5,25.7,32.8])
Exp_Votes=np.round(Exp_Votes_percent*sum(Obs_Votes)/100)
print(f"Observed Votes={Obs_Votes}")
print(f"Expected Votes={Exp_Votes}")
print("Null Hypothesis is that the sample data follows the given distribution.")
print("Alternate Hypothesis is that the sample data does not follows the given distribution.")
alpha=0.05
chi2_stats,p_value=scipy.stats.chisquare(Obs_Votes,Exp_Votes)
if p_value>alpha:
    print("Failed to reject null hypothesis for level of significance= "+str(alpha))
else:
    print("Null hypothesis is rejected for level of significance= "+str(alpha))
```

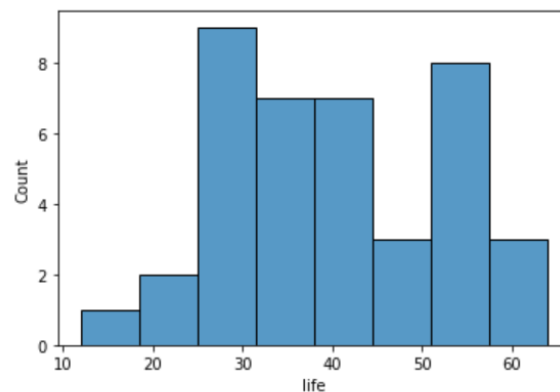
```
Observed Votes=[57 26 40]
Expected Votes=[51. 32. 40.]
Null Hypothesis is that the sample data follows the given distribution.
Alternate Hypothesis is that the sample data does not follows the given distribution.
Failed to reject null hypothesis for level of significance= 0.05
```

**Question 7:** A bulb manufacturer wants to know whether the life of the bulbs follows the normal distribution. Forty bulbs are randomly sampled, and their life, in months, are observed.

```
path = "/content/drive/MyDrive/SDS dataset/bulb_life.csv"
data = pd.read_csv(path)
data
```

|    | bulb | life |
|----|------|------|
| 0  | 1    | 31   |
| 1  | 2    | 33   |
| 2  | 3    | 34   |
| 3  | 4    | 51   |
| 4  | 5    | 24   |
| 5  | 6    | 41   |
| 6  | 7    | 58   |
| 7  | 8    | 53   |
| 8  | 9    | 27   |
| 9  | 10   | 52   |
| 10 | 11   | 40   |
| 11 | 12   | 47   |
| 12 | 13   | 37   |
| 13 | 14   | 27   |

```
import seaborn as sns
import matplotlib.pyplot as plt
sns.histplot(data=data, x='life', bins=8)
plt.show()
```



```
from scipy.stats import norm
mean=np.mean(data['life'])
std=np.std(data['life'])
bins=8
interval=[]
for i in range(1,9):
```



```
val=norm.ppf(i/bins,mean,std)
interval.append(val)
interval
```

```
[26.056476255663902,
 31.762378497044118,
 36.02928282685201,
 39.85,
 43.67071717314799,
 47.937621502955885,
 53.6435237443361,
 inf]
```

```
interval.insert(0,-np.inf)
interval
```

```
[-inf,
 26.056476255663902,
 31.762378497044118,
 36.02928282685201,
 39.85,
 43.67071717314799,
 47.937621502955885,
 53.6435237443361,
 inf]
```

```
interval.insert(0,-np.inf)
interval
```

```
[-inf,
 26.056476255663902,
 31.762378497044118,
 36.02928282685201,
 39.85,
 43.67071717314799,
 47.937621502955885,
 53.6435237443361,
 inf]
```

```
df=pd.DataFrame({'lower_limit':interval[:-1],'upper_limit':interval[1:]})
df
```

|   | lower_limit | upper_limit |
|---|-------------|-------------|
| 0 | -inf        | 26.056476   |
| 1 | 26.056476   | 31.762378   |
| 2 | 31.762378   | 36.029283   |
| 3 | 36.029283   | 39.850000   |
| 4 | 39.850000   | 43.670717   |
| 5 | 43.670717   | 47.937622   |
| 6 | 47.937622   | 53.643524   |
| 7 | 53.643524   | inf         |



```
print("Null Hypothesis is that the sample data follows normal distribution.")
print("Alternate Hypothesis is that the sample data does not follow normal distribution.")
alpha=0.05
chi2_stats,p_value=stats.chisquare(df['obs_freq'],df['exp_freq'])
print("chisquare_statistics=",chi2_stats,'and p_value',p_value )
if p_value>alpha:
    print("Failed to reject null hypothesis for level of significance= "+str(alpha))
else:
    print("Null hypothesis is rejected for level of significance= "+str(alpha))
```

Null Hypothesis is that the sample data follows normal distribution.  
 Alternate Hypothesis is that the sample data does not follow normal distribution.  
 chisquare\_statistics= 6.4 and p\_value 0.4938946499688296  
 Failed to reject null hypothesis for level of significance= 0.05

```
p=2
DOF=len(df['obs_freq'])-p-1
chi2.ppf(0.95,DOF)
```

11.070497693516351

**Question 8:** Check whether the dice is unbiased. It is tossed 90 times and the counts of outcomes are given in table.

```
path = "/content/drive/MyDrive/SDS dataset/uniform_dice.csv"
data = pd.read_csv(path)
data
```

|   | face | obs_freq |
|---|------|----------|
| 0 | 1    | 17       |
| 1 | 2    | 11       |
| 2 | 3    | 18       |
| 3 | 4    | 12       |
| 4 | 5    | 15       |
| 5 | 6    | 17       |

```
data['exp_freq']=int(sum(data['obs_freq'])/6)
print(data)
```

|   | face | obs_freq | exp_freq |
|---|------|----------|----------|
| 0 | 1    | 17       | 15       |
| 1 | 2    | 11       | 15       |
| 2 | 3    | 18       | 15       |
| 3 | 4    | 12       | 15       |
| 4 | 5    | 15       | 15       |
| 5 | 6    | 17       | 15       |

```
print("Null Hypothesis is that the sample data follows uniform distribution.")
print("Alternate Hypothesis is that the sample data does not follow uniform distribution.")
alpha=0.05
chi2_stats,p_value=stats.chisquare(data['obs_freq'],data['exp_freq'])
print("chisquare_statistics=",chi2_stats,'and p_value',p_value )
if p_value>alpha:
    print("Failed to reject null hypothesis for level of significance= "+str(alpha))
else:
    print("Null hypothesis is rejected for level of significance= "+str(alpha))
```

Null Hypothesis is that the sample data follows uniform distribution.  
 Alternate Hypothesis is that the sample data does not follow uniform distribution.  
 chisquare\_statistics= 2.8 and p\_value 0.7307864865887586  
 Failed to reject null hypothesis for level of significance= 0.05

Conclusion

Test for goodness of fit & independence of attributes has been performed for the given questions.

Signature of Faculty